

Emoji Classifier Mini-Challenge - July 16th, 2025

Goal: Train a model that predicts the correct emoji label for a short text message (or phrase).

Find/Create the Data - Example form: CSV with columns: `text`, `emoji`, `emoji_name`.

Do (minimum)

1. **Load & split data** (train/val/test; stratify by emoji).
2. **Baseline:** We need some way to take words and quantify them, some examples include: Bag-of-Words or TF-IDF Hint: You will likely need to encode your labels in some way as well
3. **Classifier:** A classifier is required to classify which words belong to which emoji - choose wisely.
4. **Improved attempt:** Change features and/or model (e.g., add n-grams, char n-grams, class weights, or try a small transformer fine-tune). Compare to baseline.
5. **Report metrics:** Top-1 accuracy + Macro-F1 on test set. Briefly note what helped.
6. **Show 5 errors:** text, true emoji, predicted emoji.

Turn In: Notebook (runs end-to-end) + short write-up (≤ 1 page) + test predictions CSV.